

Global Facilities - Design & Construction Standard - General - LEED Specification

ECN Facility:	CORP FAC	ECN Area:	ENGINEERING
Owner(s)	ENGINEERING	Target Audience:	All Facilities
Document Type:	Standard	Document ID (pdf):	A3YRXSD74VDV-57553043-459
Version:	2	Last Modified:	08/15/23
Current ECN:	101148406		

[ECN History](#)[Submit a Change Request](#)

1.0 Purpose

- 1.1 This document serves as a System Standard document that provides specifications that shall be followed during the Design, Engineering and Construction of facility systems as part of new Greenfield FAB Construction projects.
- 1.2 This Standard is applicable for all Facilities Systems.
- 1.3 The document is intended to supplement the actual Construction Drawings and Construction Specifications for Construction projects.

2.0 Scope

- 2.1 The document covers the new Basebuild System installations only and does **NOT** cover tool installation.
- 2.2 This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites for new Greenfield FAB Construction projects.

3.0 Definitions and Acronyms

Term/Acronym	Definition / Description
BD+C	Building Design and Construction
BOD	Basis of Design
CRI	Color Rendering Index
FCT	Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs
GFC	Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
GFTT	Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
Global Facilities Central Team	This is used interchangeably with Innovation & Integration Team
Global Facilities Construction Team	This is used interchangeably with Global Construction Team
Global Facilities Operations Support Team	This is used interchangeably with Global Facilities Operations Team
Global Facilities Planning Team	This is used interchangeably with Global Facilities Line Planning Team
IAQ	Indoor air quality
LEED	Leadership in Energy and Environmental Design
LID	Low-impact development
OPR	Owner's project requirements
REC	Renewable Energy Certificates
SRI	Solar Reflectance Index

4.0 Roles and Responsibilities

Roles	Responsibilities
GFTT Lead (Discipline Engineer)	• Develop and maintain System Standard document • Ensure Best Known Methods (BKM), lessons learned, and emerging technologies are documented in System Standard • Perform periodic document review and update System Standard
FCT Lead (Discipline Engineer)	• Review and provide technical feedback on System Standard • Support the Site teams by providing knowledge and technical expertise
Site Project Construction Teams	• Review and provide technical feedback on System Standard • Incorporate System Standard into Project Engineering, Design, and Construction • Document project deviations, exceptions, and exclusions from System Standard • Share new lessons learned to GFTT Lead, to determine if it is Best Known Method (BKM) that can be captured in future revision of System Standard
Site Facilities Teams	• Use System Standard as technical reference

5.0 Overview of Systems

- 5.1 LEED (Leadership in Energy and Environmental Design) is an international symbol of sustainability excellence that signifies a building is lowering carbon emissions, conserving resources, reducing operating costs, prioritizing sustainable practices and creating a healthier environment.
- 5.2 LEED is for all building types and all building phases including new construction, interior fit outs, operations and maintenance and core and shell.
- 5.3 LEED is generally organized by the following broad concepts:
- INTEGRATIVE PROCESS
 - LOCATION AND TRANSPORTATION
 - SUSTAINABLE SITES
 - WATER EFFICIENCY
 - ENERGY AND ATMOSPHERE
 - MATERIALS AND RESOURCES
 - INDOOR ENVIRONMENTAL QUALITY
 - INNOVATION
 - REGIONAL PRIORITY

- 5.4 Projects pursuing LEED certification earn points for various green building strategies across several categories based on the number of points achieved. There are four LEED rating levels available: Certified, Silver, Gold or Platinum. **Figure 1** shows the LEED points that are required for different rating levels. **Figure 2** is an example LEED scorecard which shows the details score of each item.

Chart 1: Score requirements for rating level of LEED

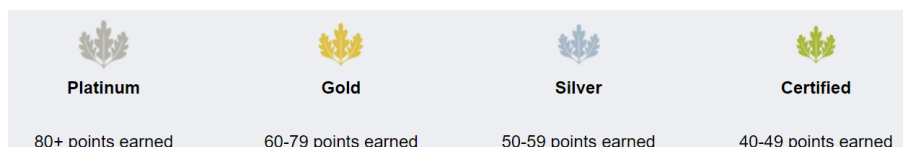


Chart 2: LEED v4 BD+C scorecard

Y ? N				Credit		Integrative Process		1	
0 0 0				Location and Transportation		16		16	
				Credit	LEED for Neighborhood Development Location	16			
				Credit	Sensitive Land Protection	1			
				Credit	High Priority Site	2			
				Credit	Surrounding Density and Diverse Uses	5			
				Credit	Access to Quality Transit	5			
				Credit	Bicycle Facilities	1			
				Credit	Reduced Parking Footprint	1			
				Credit	Green Vehicles	1			
0 0 0				Sustainable Sites		10		10	
Y				Prereq	Construction Activity Pollution Prevention	Required			
				Credit	Site Assessment	1			
				Credit	Site Development - Protect or Restore Habitat	2			
				Credit	Open Space	1			
				Credit	Rainwater Management	3			
				Credit	Heat Island Reduction	2			
				Credit	Light Pollution Reduction	1			
0 0 0				Water Efficiency		11		11	
Y				Prereq	Outdoor Water Use Reduction	Required			
Y				Prereq	Indoor Water Use Reduction	Required			
Y				Prereq	Building-Level Water Metering	Required			
				Credit	Outdoor Water Use Reduction	2			
				Credit	Indoor Water Use Reduction	6			
				Credit	Cooling Tower Water Use	2			
				Credit	Water Metering	1			
0 0 0				Energy and Atmosphere		33		33	
Y				Prereq	Fundamental Commissioning and Verification	Required			
Y				Prereq	Minimum Energy Performance	Required			
Y				Prereq	Building-Level Energy Metering	Required			
Y				Prereq	Fundamental Refrigerant Management	Required			
				Credit	Enhanced Commissioning	6			
				Credit	Optimize Energy Performance	18			
				Credit	Advanced Energy Metering	1			
				Credit	Demand Response	2			
				Credit	Renewable Energy Production	3			
				Credit	Enhanced Refrigerant Management	1			
0 0 0				Materials and Resources		13		13	
Y				Prereq	Storage and Collection of Recyclables	Required			
Y				Prereq	Construction and Demolition Waste Management Planning	Required			
				Credit	Building Life-Cycle Impact Reduction	5			
				Credit	Building Product Disclosure and Optimization - Environmental Product Declarations	2			
				Credit	Building Product Disclosure and Optimization - Sourcing of Raw Materials	2			
				Credit	Building Product Disclosure and Optimization - Material Ingredients	2			
				Credit	Construction and Demolition Waste Management	2			
0 0 0				Indoor Environmental Quality		16		16	
Y				Prereq	Minimum Indoor Air Quality Performance	Required			
Y				Prereq	Environmental Tobacco Smoke Control	Required			
				Credit	Enhanced Indoor Air Quality Strategies	2			
				Credit	Low-Emitting Materials	3			
				Credit	Construction Indoor Air Quality Management Plan	1			
				Credit	Indoor Air Quality Assessment	2			
				Credit	Thermal Comfort	1			
				Credit	Interior Lighting	2			
				Credit	Daylight	3			
				Credit	Quality Views	1			
				Credit	Acoustic Performance	1			
0 0 0				Innovation		6		6	
				Credit	Innovation	5			
				Credit	LEED Accredited Professional	1			
0 0 0				Regional Priority		4		4	
				Credit	Regional Priority: Specific Credit	1			
				Credit	Regional Priority: Specific Credit	1			
				Credit	Regional Priority: Specific Credit	1			
				Credit	Regional Priority: Specific Credit	1			
0 0 0				TOTALS		Possible Points: 110		110	
Certified: 40 to 49 points, Silver: 50 to 59 points, Gold: 60 to 79 points, Platinum: 80 to 110									

- 5.5 This standard covers only new Greenfield FAB Construction projects. All new Micron Greenfield FAB Construction Projects are required to achieve minimum LEED Gold certification.

6.0 LEED Gold Approach – Approach for new Greenfield FAB Construction projects

- 6.1 For all new Greenfield FAB Construction projects, the requirement to achieve LEED Gold certification shall be highlighted at the beginning of the project. This includes, but is not limited to, during the early project scenario planning, exploratory discussions, and Concept Design stages of the project.
- 6.2 A LEED certified consultant shall be engaged and contracted during the project Detailed Design Stage, to perform the LEED feasibility study, provide engineering solutions and recommendations to achieve the required LEED points, and provide the formal LEED application submission.
- 6.3 A LEED feasibility study shall be performed at the beginning of Design Stage for the project, for preliminary evaluation of how feasible it is for the project to achieve the minimum LEED Gold certification.
 - 6.3.1 The items that are required in order to meet the LEED Gold certification shall be identified and included in the project Design.
 - 6.3.2 The LEED feasibility study shall be updated throughout the Design process to keep accurate records of LEED items and points achievable.
 - 6.3.3 The LEED study shall also include an evaluation of the points required to achieve LEED Platinum certification as a stretch target, and whether it is feasible.
- 6.4 The following section is a summary outline of the LEED BD+C scorecard including the categories and scoring.
 - The points listed in category “Y” are points that are highly likely to be able to achieve, and should/must be included in the project; the sum is **62**.
 - The points listed in category “?” are points that may be achievable or questionable to achieve for a project, depending on site location/condition or it may need to take a lot of effort to achieve; the sum is **40**.
 - The points listed in category “N” are points that are highly unlikely to be able to achieve; the sum is **18**.
 - “Credit” means that possible LEED points are achievable for this item.
 - “Pre-Requisite” means this item must be completed to qualify for LEED.

According to the summary outline estimation, the suggested points achievable (“Y”) is **62**. Typical recommendation to achieve LEED Gold is to target minimum of 65 points during the application submission. Therefore, a minimum of additional 3 points shall be considered from site location/condition related items, when the site location is confirmed. If these points are not possible, then the other items shall be checked for feasibility.

Y	?	N	Credit/ Prerequisite	Category / item	Intent
1			Credit	Integrative Process	To support high-performance, cost-effective project outcomes through an early analysis of the interrelationships among systems.
7	8	1		Location and Transportation	Total points: 16
		12	Credit	LEED for Neighborhood Development Location	To avoid development on inappropriate sites. To reduce vehicle distance traveled. To enhance livability and improve human health by encouraging daily physical activity.
1			Credit	Sensitive Land Protection	To avoid the development of environmentally sensitive lands and reduce the environmental impact from the location of a building on a site.
1		1	Credit	High Priority Site	To encourage project location in areas with development constraints and promote the health of the surrounding area.
2	3		Credit	Surrounding Density and Diverse Uses	To conserve land and protect farmland and wildlife habitat by encouraging development in areas with existing infrastructure. To promote walkability, and transportation efficiency and reduce vehicle distance traveled. To improve public health by encouraging daily physical activity.
	5		Credit	Access to Quality Transit	To encourage development in locations shown to have multimodal transportation choices or otherwise reduced motor vehicle use, thereby reducing greenhouse gas emissions, air pollution, and other environmental and public health harms associated with motor vehicle use.
1			Credit	Bicycle Facilities	To promote bicycling and transportation efficiency and reduce vehicle distance traveled. To improve public health by encouraging utilitarian and recreational physical activity.
1			Credit	Reduced Parking Footprint	To minimize the environmental harms associated with parking facilities, including automobile dependence, land consumption, and rainwater runoff.
1			Credit	Green Vehicles	To reduce pollution by promoting alternatives to conventionally fueled automobiles

Y	?	N	Credit/ Prerequisite	Category / item	Intent
8	2			SUSTAINABLE SITES	Total points: 10
			Prerequisite	Construction Activity Pollution Prevention	To reduce pollution from construction activities by controlling soil erosion, waterway sedimentation, and airborne dust.
1			Credit	Site Assessment	To assess site conditions before design to evaluate sustainable options and inform related decisions about site design.
1	1		Credit	Site Development - Protect or Restore Habitat	To conserve existing natural areas and restore damaged areas to provide habitat and promote biodiversity.
	1		Credit	Open Space	To create exterior open space that encourages interaction with the environment, social interaction, passive recreation, and physical activities.
3			Credit	Rainwater Management	To reduce runoff volume and improve water quality by replicating the natural hydrology and water balance of the site, based on historical conditions and undeveloped ecosystems in the region.
2			Credit	Heat Island Reduction	To minimize effects on microclimates and human and wildlife habitats by reducing heat islands. conditions and undeveloped ecosystems in the region.
1			Credit	Light Pollution Reduction	To minimize effects on microclimates and human and wildlife habitats by reducing heat islands. conditions and undeveloped ecosystems in the region.
11				Water Efficiency	Total points: 11
			Prerequisite	Outdoor Water Use Reduction	To reduce outdoor water consumption.
			Prerequisite	Indoor Water Use Reduction	To reduce indoor water consumption.
			Prerequisite	Building-Level Water Metering	To support water management and identify opportunities for additional water savings by tracking water consumption.
2			Credit	Outdoor Water Use Reduction	To reduce outdoor water consumption.
6			Credit	Indoor Water Use Reduction	To reduce indoor water consumption.
2			Credit	Cooling Tower Water Use	To conserve water used for cooling tower makeup while controlling microbes, corrosion, and scale in the condenser water system.
1			Credit	Water Metering	To support water management and identify opportunities for additional water savings by tracking water consumption.

Y	?	N	Credit/ Prerequisite	Category / item	Intent
13	20	10		Energy and Atmosphere	Total points: 33
			Prerequisite	Fundamental Commissioning and Verification	To support the design, construction, and eventual operation of a project that meets the owner's project requirements for energy, water, indoor environmental quality, and durability.
			Prerequisite	Minimum Energy Performance	To reduce the environmental and economic harms of excessive energy use by achieving a minimum level of energy efficiency for the building and its systems.
			Prerequisite	Building-Level Energy Metering	To support energy management and identify opportunities for additional energy savings by tracking building-level energy use.
			Prerequisite	Fundamental Refrigerant Management	To reduce stratospheric ozone depletion.
4	2		Credit	Enhanced Commissioning	To further support the design, construction, and eventual operation of a project that meets the owner's project requirements for energy, water, indoor environmental quality, and durability.
6	2	10	Credit	Optimize Energy Performance	To achieve increasing levels of energy performance beyond the prerequisite standard to reduce environmental and economic harms associated with excessive energy use.
1			Credit	Advanced Energy Metering	To support energy management and identify opportunities for additional energy savings by tracking building-level and system-level energy use.
1	1		Credit	Demand Response	To increase participation in demand response technologies and programs that make energy generation and distribution systems more efficient, increase grid reliability, and reduce greenhouse gas emissions.
	3		Credit	Renewable Energy Production	To reduce the environmental and economic harms associated with fossil fuel energy by increasing self-supply of renewable energy.
1			Credit	Enhanced Refrigerant Management	To reduce ozone depletion and support early compliance with the Montreal Protocol while minimizing direct contributions to climate change.
	2		Credit	Green Power and Carbon Offsets	To encourage the reduction of greenhouse gas emissions through the use of grid-source, renewable energy technologies and carbon mitigation projects.

Y	?	N	Credit/ Prerequisite	Category / item	Intent
6	7			Materials and Resources	Total points: 13
			Prereq	Storage and Collection of Recyclables	To reduce the waste that is generated by building occupants and hauled to and disposed of in landfills.
			Prereq	Construction and Demolition Waste Management Planning	To reduce construction and demolition waste disposed of in landfills and incineration facilities by recovering, reusing, and recycling materials.
1	4		Credit	Building Life-Cycle Impact Reduction	To encourage adaptive reuse and optimize the environmental performance of products and materials.
1	1		Credit	Building Product Disclosure and Optimization - Environmental Product Declarations	To encourage the use of products and materials for which life-cycle information is available and that have environmentally, economically, and socially preferable life-cycle impacts. To reward project teams for selecting products from manufacturers who have verified improved environmental life-cycle impacts.
1	1		Credit	Building Product Disclosure and Optimization - Sourcing of Raw Materials	To encourage the use of products and materials for which life cycle information is available and that have environmentally, economically, and socially preferable life cycle impacts. To reward project teams for selecting products verified to have been extracted or sourced in a responsible manner.
1	1		Credit	Building Product Disclosure and Optimization - Material Ingredients	To encourage the use of products and materials for which life-cycle information is available and that have environmentally, economically, and socially preferable life-cycle impacts. To reward project teams for selecting products for which the chemical ingredients in the product are inventoried using an accepted methodology and for selecting products verified to minimize the use and generation of harmful substances. To reward raw material manufacturers who produce products verified to have improved life-cycle impacts.
2			Credit	Construction and Demolition Waste Management	To reduce construction and demolition waste disposed of in landfills and incineration facilities by recovering, reusing, and recycling materials

Y	?	N	Credit/ Prerequisite	Category / item	Intent
6	3	7		Indoor Environmental Quality	Total points: 16
			Prereq	Minimum Indoor Air Quality Performance	To contribute to the comfort and well-being of building occupants by establishing minimum standards for indoor air quality (IAQ).
			Prereq	Environmental Tobacco Smoke Control	To prevent or minimize exposure of building occupants, indoor surfaces, and ventilation air distribution systems to environmental tobacco smoke.
2			Credit	Enhanced Indoor Air Quality Strategies	To promote occupants' comfort, well-being, and productivity by improving indoor air quality.
1	1	1	Credit	Low-Emitting Materials	To reduce concentrations of chemical contaminants that can damage air quality, human health, productivity, and the environment.
1			Credit	Construction Indoor Air Quality Management Plan	To promote the well-being of construction workers and building occupants by minimizing indoor air quality problems associated with construction and renovation.
1	1		Credit	Indoor Air Quality Assessment	To establish better quality indoor air in the building after construction and during occupancy.
		1	Credit	Thermal Comfort	To promote occupants' productivity, comfort, and well-being by providing quality thermal comfort.
1	1		Credit	Interior Lighting	To promote occupants' productivity, comfort, and well-being by providing high-quality lighting.
		3	Credit	Daylight	To connect building occupants with the outdoors, reinforce circadian rhythms, and reduce the use of electrical lighting by introducing daylight into the space.
		1	Credit	Quality Views	To give building occupants a connection to the natural outdoor environment by providing quality views.
		1	Credit	Acoustic Performance	To provide workspaces and classrooms that promote occupants' well-being, productivity, and communications through effective acoustic design.

Y	?	N	Credit/ Prerequisite	Category / item	Intent
6				Innovation	Total points: 6
5			Credit	Innovation	To encourage projects to achieve exceptional or innovative performance.
1			Credit	LEED Accredited Professional	To encourage the team integration required by a LEED project and to streamline the application and certification process.
4				Regional Priority	Total points: 4
4			Credit	Regional Priority: Specific Credit	To provide an incentive for the achievement of credits that address geographically specific environmental, social equity, and public health priorities.
TOTAL POINTS					
Y	?	N	-		
62	40	18	-		

7.0 LEED Gold Strategy - Details and Recommendations to obtain LEED Points

The following section is a detailed summary of requirements for the LEED points, and the recommendations of methods to obtain the LEED points. For more details and specifics, refer to the **LEED v4 for BUILDING DESIGN AND CONSTRUCTION (BD+C) material**.

7.1 INTEGRATIVE PROCESS: (Total: 1 point, suggested: 1 point)

Credit/ Prerequisite	Points	Requirement
	Y/?/N	
Credit	1/0/0	Beginning in pre-design and continuing throughout the design phases, identify and use opportunities to achieve synergies across disciplines and building systems. Use the analyses described below to inform the owner project requirements (OPR), basis of design (BOD), design documents, and construction documents. v4.0 Conducting Energy-Related and Water-Related Systems analyses described in LEED standard. <i>If above is difficult to execute. Recommend go for v4.1 as it is more flexible.</i> v4.1 Choose two of the following to analyze: Energy-Related Systems / Water-Related Systems / Site Selection / Social Equity / Health & Well-being

Location and Transportation: (Total: 16 points, suggested: 8 point)

Credit/ Prerequisite	Points	Requirement
	Y/?/N	
7.2.1 LEED for Neighborhood Development Location		
Credit	0/0/12	Locate the project within the boundary of a development certified under LEED for Neighborhood Development. (12) It depends on site condition.
7.2.2 Sensitive Land Protection		
Credit	1/0/0	Option 1. (1) Locate the development footprint on land that has been previously developed. OR Option 2. (1) - Recommend Locate the development footprint on land that has been previously developed or that does not meet the criteria list in the LEED V4 standard for sensitive land. Recommend Option 2 since the site will not be in sensitive land normally.
7.2.3 High Priority Site v4.1 High-Priority Site and Equitable Development		
Credit	1/0/1	Option 1. Historic District (1) OR Option 2. Priority Designation (1) - Recommend OR Option 3. Brownfield Remediation (2) Recommend Option 2 since our Fab normally site in national priority list. If above is difficult to execute. Recommend go for v4.1 as it is more flexible. v4.1 Option 1. High Priority Site Path 1. Economically Disadvantaged Community Location (1) OR Path 2. Brownfield Remediation (1) AND/OR Option 2. Equitable Development Path 1. Equity & Community Benefits (1) - Recommend OR Path 2. Affordable Housing in Residential or Mixed-Use Projects (1)
7.2.4 Surrounding Density and Diverse Uses		
Credit	2/3/0	Option 1. Surrounding Density (3) (depends on site condition) Locate on a site whose surrounding existing density within a ¼-mile (400-meter) radius of the project boundary meets the values in LEED V4. AND/OR Option 2. Diverse Uses (2) - Recommend

Credit/ Prerequisite	Points Y/?/N	Requirement																												
		Construct or renovate a building or a space within a building such that the building’s main entrance is within a ½-mile (800-meter) walking distance of the main entrance of four to seven (1 point) or eight or more (2 points) existing and publicly available diverse uses. Recommend set up at least 4 publicly diverse uses to get 1 point for option 2 since option 1 depends on site condition. Ex. Convenience store, Restaurant, Gym, Bank and so on. If above is difficult to execute. Recommend go for v4.1 as it provides additional option Option 3. Walkable Location Locate on a site with a Walk Score® or equivalent third-party walkability assessment for the following thresholds, as listed below. <table><tr><th>Walk Score*</th><th>Points</th><th>Points (Core & Shell)</th><th>Points (Healthcare)</th></tr><tr><td>90 to 100</td><td>5</td><td>6</td><td>-</td></tr><tr><td>80 to 89</td><td>4</td><td>4</td><td>-</td></tr><tr><td>70 to 79</td><td>3</td><td>3</td><td>-</td></tr><tr><td>60 to 69</td><td>2</td><td>2</td><td>-</td></tr><tr><td>50 to 59</td><td>1</td><td>1</td><td>-</td></tr><tr><td>≥ 50</td><td>-</td><td>-</td><td>1</td></tr></table> Get Your Walk Score	Walk Score*	Points	Points (Core & Shell)	Points (Healthcare)	90 to 100	5	6	-	80 to 89	4	4	-	70 to 79	3	3	-	60 to 69	2	2	-	50 to 59	1	1	-	≥ 50	-	-	1
Walk Score*	Points	Points (Core & Shell)	Points (Healthcare)																											
90 to 100	5	6	-																											
80 to 89	4	4	-																											
70 to 79	3	3	-																											
60 to 69	2	2	-																											
50 to 59	1	1	-																											
≥ 50	-	-	1																											
7.2.5 Access to Quality Transit (depends on site condition)																														
Credit	0/5/0	Locate any functional entry of the project within a ¼-mile (400-meter) walking distance of existing or planned bus, streetcar, or rideshare stops, or within a ½-mile (800-meter) walking distance of existing or planned bus rapid transit stops, light or heavy rail stations, commuter rail stations, or commuter ferry terminals. If above is difficult to execute. Recommend go for v4.1 as it provides additional option Path 2. Access to Project-sponsored Transit Service Commit to providing year-round transit service (vans, shuttles, buses) for regular occupants and visitors that meets the minimums listed in Table below, Service must be guaranteed for at least 3 years from the project’s certificate of occupancy. <table><tr><th>Total daily trips</th><th>Points</th></tr><tr><td>30</td><td>1</td></tr><tr><td>45</td><td>2</td></tr></table>	Total daily trips	Points	30	1	45	2																						
Total daily trips	Points																													
30	1																													
45	2																													

Credit/ Prerequisite	Points	Requirement
	Y/?/N	
7.2.6 Bicycle Facilities		
Credit	1/0/0	Recommend Design or locate the bicycle network, Bicycle Storage and Shower Rooms according to the request of LEED V4.
7.2.7 Reduced Parking Footprint		
Credit	1/0/0	Recommend Do not exceed the minimum local code requirements for parking capacity. If above is difficult to execute. Recommend go for v4.1 as it provides additional option Option 2. Carshare (1 pt) - Recommend Provide dedicated parking for carshare vehicles. Provide carshare vehicle parking space(s) for at least 1% of total parking spaces, rounded up. If the project has fewer than 100 parking spaces, provide one carshare vehicle parking space. OR Option 3. Unbundling Parking (1 pt)
7.2.8 Green Vehicles		
v4.1 Electric Vehicles		
Credit	1/0/0	Designate 5% of all parking spaces used by the project as preferred parking for green vehicles. In addition to preferred parking for green vehicles, meet one of the following two options for alternative-fuel fueling stations: at least 2% of all parking spaces Option 1. Electric Vehicle Charging (1) Or Option 2. Liquid, gas, or battery facilities (1) Recommend reserving parking lot for green vehicles and design Electric Vehicle Charging station. If above is difficult to execute. Recommend go for v4.1 as it provides additional option Option 2. Electric Vehicle Ready Infrastructure (1 pt) Make 10% of parking spaces or at least 6 spaces, whichever is greater, EV Ready.

7.2 Sustainable Sites: (Total: 10 points, suggested: 2 points)

Credit/ Prerequisite	Points Y/?/N	Requirement
7.3.1 Construction Activity Pollution Prevention		
Prerequisite		Must: Create and implement an erosion and sedimentation control plan for all construction activities associated with the project.
7.3.2 Site Assessment		
Credit	1/0/0	Recommend Complete and document a site survey or assessment that includes list in LEED V4.
7.3.3 Site Development - Protect or Restore Habitat (depends on site condition)		
Credit	1/1/0	Preserve and protect from all development and construction activity 40% of the greenfield area on the site (if such areas exist). AND Option 1. On-Site Restoration (2) Or Option 2. Financial Support (1) IF site condition allowed, Recommend option 1 by using native or adapted vegetation, restore 15% (including the building footprint) of all portions of the site identified as previously disturbed. If option 1 fails, then go for in option 2. Option 1 in v4.1 is easier to achieve as the restoration area requirement was reduced from a minimum of 30% to 15%, and a second threshold was added at 25% for two points to reward projects for increased restoration.
7.3.4 Open Space (depends on site condition)		
Credit	0/1/0	IF site condition allowed, Recommend Provide outdoor space greater than or equal to 30% of the total site area (including building footprint). A minimum of 25% of that outdoor space must be vegetated (turf grass does not count as vegetation) or have <i>overhead vegetated canopy</i> .
7.3.5 Rainwater Management		
Credit	3/0/0	Option 1. Percentile of Rainfall Events Using <i>low-impact development (LID)</i> and <i>green infrastructure</i> to manage on site the runoff from the developed site. Path 1. 95th Percentile (2) OR Path 2. 98th Percentile (3) Recommend OR Path 3. Zero Lot Line projects only – 85th Percentile (3) OR Option 2. Natural Land Cover Conditions (3)

Credit/ Prerequisite	Points	Requirement																
	Y/?/N																	
		Manage on site the annual increase in runoff volume from the natural land cover condition to the postdeveloped condition. Recommend design rain recover tank and LID/GI to meet option 1. Consider v4.1 as it is now able to earn points for retaining runoff for the 80th, 85th, or 90th percentile of regional or local rainfall events, whereas the credit previously required the 95th or 98th percentile. <table><tr><td>All Projects</td><td>Zero lot line Projects</td><td>Points</td><td>Points Healthcare</td></tr><tr><td>80th Percentile</td><td>70th Percentile</td><td>1</td><td>1</td></tr><tr><td>85th Percentile</td><td>75th Percentile</td><td>2</td><td>2</td></tr><tr><td>90th Percentile</td><td>80th Percentile</td><td>3</td><td>-</td></tr></table>	All Projects	Zero lot line Projects	Points	Points Healthcare	80 th Percentile	70 th Percentile	1	1	85 th Percentile	75 th Percentile	2	2	90 th Percentile	80 th Percentile	3	-
All Projects	Zero lot line Projects	Points	Points Healthcare															
80 th Percentile	70 th Percentile	1	1															
85 th Percentile	75 th Percentile	2	2															
90 th Percentile	80 th Percentile	3	-															
7.3.6 Heat Island Reduction																		
Credit	2/0/0	Option 1. Nonroof and Roof (2) (Recommend) Through applying Nonroof Measures, High-Reflectance Roof and Vegetated Roof to achieve Area of Nonroof Measures 0.5 + Area of High-Reflectance Roof 0.75 + Area of Vegetated Roof 0.75 ≥ Total Site Paving Area + Total Roof Area OR Option 2. Parking under Cover (1) Place a minimum of 75% of <i>parking spaces under cover</i>. Recommend achieve option 1 by designing open-grid pavement system, High solar reflectance index (SRI) roofing, solar panel, vegetated roof and so on.																
7.3.7 Light Pollution Reduction																		
Credit	1/0/0	Meet uplight and light trespass requirements, using either the backlight-uplight-glare (BUG) method (Option 1) or the calculation method (Option 2). Recommend designing exterior luminaires to meet the requirements.																

7.3 Water Efficiency: (Total: 11 points, suggested: 11 points)

Credit/ Prerequisite	Point	Requirement
	Y/?/N	
7.4.1 Outdoor Water Use Reduction		
Prerequisite		Must: Reduce outdoor water use through one of the following options. Option 1. No Irrigation Required OR Option 2. Reduced Irrigation Reduce the project’s landscape water requirement by at least 30% from the calculated baseline for the site’s peak watering month. Recommend xerophyte plant and high efficiency irrigation system.
7.4.2 Indoor Water Use Reduction		
Prerequisite		Must: Building Water Use For the fixtures and fittings listed in LEED standard, as applicable to the project scope, reduce aggregate water consumption by 20% from the baseline. Appliance and Process Water Use Install appliances, equipment, and processes within the project scope that meet the requirements listed in LEED standard. Recommend 1. Building fixtures and fittings, likes toilets, urinals, private lavatory faucets, etc. are eligible for labeling must be WaterSense labeled (or a local equivalent for projects outside the U.S.). 2. Install appliances, equipment, and processes list in the standard meet ENERGY STAR or performance equivalent. 3. Cooling tower equip with makeup water meters, conductivity controllers and overflow alarms. Installing drift eliminators that reduce drift to maximum of 0.002% of recirculated water volume for counterflow towers and 0.005% of recirculated water flow for cross-flow towers.
7.4.3 Building-Level Water Metering		
Prerequisite		Must: Install permanent water meters that measure the total potable water use for the building and associated grounds. Meter data must be compiled into monthly and annual summaries; meter readings can be manual or automated. Recommend: Government water meter is applicable.

Credit/ Prerequisite	Point Y/?/N	Requirement												
7.4.4 Outdoor Water Use Reduction														
Credit	2/0/0	Reduce outdoor water use through one of the following options. Option 1. No Irrigation Required (2) OR Option 2. Reduced Irrigation (1 for 50%, 2 for 100% reduction) (Recommend) Reduce the project’s landscape water requirement by at least 50% from the calculated baseline for the site’s peak watering month. Recommend installing water recovery system or rain recovery tank and recovery water must use in irrigation.												
7.4.5 Indoor Water Use Reduction														
Credit	6/0/0	Recommend Further reduce fixture and fitting water (50%) use from the calculated baseline in WE Prerequisite Indoor Water Use Reduction. Additional potable water savings can be earned above the prerequisite level using alternative water sources. Recommend using recovery rain water use in toilet and urinal fixtures.												
7.4.6 Cooling Tower Water Use														
v4.1 Optimize Process Water Use														
Credit	2/0/0	Conduct a one-time potable water analysis, measuring at least the five control parameters listed in Table 1. Table 1. Maximum concentrations for parameters in condenser water <table><tr><td>Parameter</td><td>Maximum level</td></tr><tr><td>Ca (as CaCO₃)</td><td>1000 ppm</td></tr><tr><td>Total alkalinity</td><td>1000 ppm</td></tr><tr><td>SiO₂</td><td>100 ppm</td></tr><tr><td>Cl⁻</td><td>250 ppm</td></tr><tr><td>Conductivity</td><td>2000 μS/cm</td></tr></table> ppm = parts per million μS/cm = micro siemens per centimeter Points for cooling tower cycles Maximum number of cycles achieved without exceeding any filtration levels or affecting operation of condenser water system (up to maximum of 10 cycles) (1) Recommend Achieve a minimum 10 cycles by increasing the level of treatment in condenser or make-up water (2) OR Meet the minimum number of cycles to earn 1 point and use a minimum 20% recycled nonpotable water (2) v4.1 provides 2 other options to achieve this credit.	Parameter	Maximum level	Ca (as CaCO ₃)	1000 ppm	Total alkalinity	1000 ppm	SiO ₂	100 ppm	Cl ⁻	250 ppm	Conductivity	2000 μS/cm
Parameter	Maximum level													
Ca (as CaCO ₃)	1000 ppm													
Total alkalinity	1000 ppm													
SiO ₂	100 ppm													
Cl ⁻	250 ppm													
Conductivity	2000 μS/cm													

Credit/ Prerequisite	Point	Requirement
	Y/?/N	
		Option 2. Optimize Water Use for Cooling (2) To be eligible for Option 2, the baseline system designated for the building using ASHRAE 90.1-2016 Appendix G Table G3.1.1 must include a cooling tower (systems 7, 8, 11, 12, and 13) OR Option 3. Process Water Use (2) Demonstrate that the project is using minimum 20% recycled alternative water to meet process water demand for 1 point, or using minimum 30% recycled alternative water to meet process water demand for 2 points. Ensure that recycled alternative water is of sufficient quality for its intended end use. Process water uses eligible for achievement of Option 3 must represent at least 10% of total building regulated water use and may not include water used for cooling. Eligible subsystems may include: • Boilers • Humidification systems • Other subsystems using process water

7.4.6 Water metering

Credit	1/0/0	Recommend Install permanent water meters for two or more of the following water subsystems, as applicable to the project: • Irrigation. • Indoor plumbing fixtures and fittings. • Domestic hot water. • Boiler with aggregate projected annual water use of 100,000 gallons (378 500 liters) or more, or boiler of more than 500,000 BtuH (150 kW). • Reclaimed water. • Other process water.
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7.4 Energy and Atmosphere: (Total: 43 points, suggested: 13 points)

Credit/ Prerequisite	Points	Requirement
	Y/?/N	
7.5.1 Fundamental Commissioning and Verification		
Prerequisite		Must: Commissioning Process Scope / Commissioning Authority Current Facilities Requirements and Operations and Maintenance Plan must follow the requirement in LEED Standard. Recommend Complete commissioning (Cx) process activities for mechanical, electrical, plumbing, and renewable energy systems and assemblies, in accordance with ASHRAE Guideline 0-2005 and ASHRAE Guideline 1.1–2007 for HVAC&R Systems.

Credit/ Prerequisite	Points	Requirement
	Y/?/N	
7.5.2 Minimum Energy Performance		
Prerequisite		Must: Option 1. Whole-Building Energy Simulation - Recommend Demonstrate an improvement of 5% for new construction projects in the proposed building performance rating compared with the baseline building performance rating. OR Option 2. Prescriptive Compliance: ASHRAE 50% Advanced Energy Design Guide Option 3. Prescriptive Compliance: Advanced Buildings™ Core Performance™ Guide Recommend design must be compliant with the mandatory provisions of ANSI/ASHRAE/IESNA Standard 90.1–2010, with errata (or a USGBC-approved equivalent standard for projects outside the U.S.) Recommend to apply option 1.
7.5.3 Building-Level Energy Metering		
Prerequisite		Must: Install new or use existing building-level energy meters, or submeters that can be aggregated to provide building-level data representing total building energy consumption (electricity, natural gas, chilled water, steam, fuel oil, propane, biomass, etc.). Recommend that Utility-owned meters capable of aggregating building-level resource use are acceptable.
7.5.4 Fundamental Refrigerant Management		
Prerequisite		Must: Do not use chlorofluorocarbon (CFC)-based refrigerants in new heating, ventilating, air-conditioning, and refrigeration (HVAC&R) systems. When reusing existing HVAC&R equipment, complete a comprehensive CFC phase-out conversion before project completion. Phase-out plans extending beyond the project completion date will be considered on their merits. Recommend using non-CFC refrigerants.
7.5.5 Enhanced Commissioning		
Credit	4/2/0	Implement commissioning process activities in addition to those required under EA Prerequisite Fundamental Commissioning and Verification. Option 1. Enhanced Systems Commissioning (3-4 points) Path 1: Enhanced Commissioning (3 points) OR Path 2: Enhanced and Monitoring-Based Commissioning (4 points) - Recommend

Credit/ Prerequisite	Points Y/?/N	Requirement
		Achieve Path 1. AND Develop monitoring-based procedures and identify points to be measured and evaluated to assess performance of energy- and water-consuming systems. AND/OR Option 2. Envelope Commissioning (2 points) Recommend • Complete commissioning process (CxP) activities for mechanical, electrical, plumbing, and renewable energy systems and assemblies in accordance with ASHRAE Guideline 0–2005 and ASHRAE Guideline 1.1–2007 for HVAC&R systems and Develop monitoring-based procedures. • If possible, Complete the following commissioning process (CxP) activities for the building's thermal envelope in accordance with ASHRAE Guideline 0–2005 and the National Institute of Building Sciences (NIBS) Guideline 3–2012, Exterior Enclosure Technical Requirements for the Commissioning Process.
7.5.6 Optimize Energy Performance		
Credit	6/0/12	Option 1. Whole-Building Energy Simulation (1–18 points) - Recommend Demonstrate an improvement of 16% at least for new construction projects in the proposed building performance rating compared with the baseline building performance rating to get 6 points. OR Option 2. Prescriptive Compliance: ASHRAE Advanced Energy Design Guide (1–6 points) Recommend go for option 1. Only the office, retail, school, or hospital building type can apply option 2.
7.5.5 Advanced Energy Metering		
Credit	1/0/0	Recommend Install <i>advanced energy metering</i> for the following: • all whole-building energy sources used by the building; and • any individual energy end uses that represent 10% or more of the total annual consumption of the building.
7.5.6 Demand Response		
v4.1 Grid Harmonization		
Credit	1/1/0	Design building and equipment for participation in demand response programs through load shedding or shifting. On-site electricity generation does not meet the intent of this credit. Case 1. Demand Response Program Available (2)

Credit/ Prerequisite	Points Y/?/N	Requirement															
		Case 2. Demand Response Program Not Available (1) - Recommend Provide infrastructure to take advantage of future demand response programs or dynamic, real-time pricing programs and complete the following activities. Recommend where demand response program is not available develop a shedding plan to achieve case 2. v4.1 provides 3rd option to achieve this credit. Case 3. Load Flexibility and Management Strategies (2) Implement one or more of the load flexibility and management strategies described below for a maximum of up to two points. All projects must install interval recording meters and have equipment capable of accepting an external signal. Load Flexibility and Management Strategies: ● Peak Load Optimization: demonstrate that strategy reduces on-peak load by at least 10% as compared to peak electrical demand referenced to the ASHRAE 90.1-2016 compliant case (1 point) ● Flexible Operating Scenarios: demonstrate that strategy moves at least 10% of peak load by a time period of 2 hours (1 point) ● On-site thermal and/or electricity storage: demonstrate that strategy reduces on-peak load by at least 10% as compared to peak electrical demand (1 point) ● Grid resilience technologies: project served by utilities with resilience programs in place, which leverage strategies such as islanding and part-load operation, automatically achieve this credit (1 point)															
7.5.7 Renewable Energy Production v4.1 Renewable Energy (combined v4.0 Renewable Energy Production and Green Power and Carbon Offsets)																	
Credit	0/3/0	Better to have (Check with Renewable team about the strategy) Use renewable energy systems to offset building energy costs. Table 1. Points for renewable energy <table> <tr> <th>Percentage renewable energy</th><th>Points (except CS)</th><th>Points (CS)</th></tr> <tr> <td>1%</td><td>1</td><td>1</td></tr> <tr> <td>3%</td><td>—</td><td>2</td></tr> <tr> <td>5%</td><td>2</td><td>3</td></tr> <tr> <td>10%</td><td>3</td><td>—</td></tr> </table> v4.1 combined v4.0 Renewable Energy Production and Green Power and Carbon Offsets provide more flexible option to achieve. Choose one or more strategies for renewable energy procurement from the categories below. Points achieved in each category may be added for a total of 5 points. ● Tier 1: On-site renewable energy generation ■ On-site renewable energy generation, environmental attributes (e.g. RECs) retained	Percentage renewable energy	Points (except CS)	Points (CS)	1%	1	1	3%	—	2	5%	2	3	10%	3	—
Percentage renewable energy	Points (except CS)	Points (CS)															
1%	1	1															
3%	—	2															
5%	2	3															
10%	3	—															

Credit/ Prerequisite	Points	Requirement
	Y/?/N	
		</

7.5 Materials and Resources: (Total: 13 points, suggested: 6 points)

Credit/ Prerequisite	Points	Requirement
	Y/?/N	
7.6.1 Storage and Collection of Recyclables		
Prerequisite		Must: Provide dedicated areas accessible to waste haulers and building occupants for the collection and storage of recyclable materials for the entire building.
7.6.2 Construction and Demolition Waste Management Planning		
Prerequisite		Must: Develop and implement a construction and demolition waste management plan which should include through waste prevention and by reusing, recovering, and recycling (RRR) of materials, contractor shall provide a final waste management report detailing all waste generated. At a minimum, Contractor shall achieve a total construction and demolition waste RRR of minimum 80%, via diversion of total construction and demolition materials from landfills and/or incineration facilities. The preferred construction and demolition waste RRR target is 81-85%.
7.6.3 Building Life-Cycle Impact Reduction		
Credit	1/4/0	Demonstrate reduced environmental effects during initial project decision-making by reusing existing building resources or demonstrating a reduction in materials use through life-cycle assessment. Achieve one of the following options. Option 1. Historic Building Reuse (5) OR Option 2. Renovation of Abandoned or Blighted Building (5) OR Option 3. Building and Material Reuse (4) OR Option 4. Whole-Building Life-Cycle Assessment (3) - Recommend Recommend reusing or using recycled content building materials from offsite, ex. concrete, steel to meet option 4. Consider v4.1 Option 2. Whole-Building Life-Cycle Assessment as it has an entry pathway that rewards the effort to conduct a whole building life-cycle assessment without having to demonstrate specific impact reductions.
7.6.4 Building Product Disclosure and Optimization - Environmental Product Declarations v4.1 Environmental Product Declarations		
Credit	1/1/0	Option 1. Environmental Product Declaration (EPD) (1) - Recommend AND/OR Option 2. Multi-Attribute Optimization (1) Recommend Use at least 20 different permanently installed products sourced from at least five different manufacturers that meet Environmental Product Declarations (EPD).

Credit/ Prerequisite	Points Y/?/N	Requirement
7.6.5 Building Product Disclosure and Optimization - Sourcing of Raw Materials v4.1 Sourcing of Raw Materials		
Credit	1/1/0	Option 1. Raw Material Source and Extraction Reporting (1) AND/OR Option 2. Leadership Extraction Practices (1) Recommend using concrete, steel and others which included recycled content to meeting option 2 requirement. As USGBC keep only option 2 in v4.1, it worth to review if it meets the requirement of 2 pts. Use products sourced from at least three different manufacturers that meet at least one of the responsible sourcing and extraction criteria below for at least 15%, by cost, of the total value of permanently installed building products in the project (1 point). Use products sourced from at least five different manufacturers that meet at least one of the responsible sourcing and extraction criteria below for at least 30%, by cost, of the total value of permanently installed building products in the project (2 points).
7.6.6 Building Product Disclosure and Optimization - Material Ingredients v4.1 Material Ingredients		
Credit	1/1/0	Option 1. Material Ingredient Reporting (1) - Recommend AND/OR Option 2: Material Ingredient Optimization (1) AND/OR Option 3: Product Manufacturer Supply Chain Optimization (1) Recommend Use at least 20 different permanently installed products sourced from at least five different manufacturers that meet 1.Health Product Declaration (HPD) 2.Cradle to Cradle (C2C) 3.ANSI/BIFMA e3 Furniture Sustainability Standard 4.Product Lens Certification 5.Facts - NSF/ANSI 336: Sustainability Assessment for Commercial Furnishings Fabric at any certification level.
7.6.7 Construction and Demolition Waste Management		
Credit	2/0/0	Recycle and/or salvage nonhazardous construction and demolition materials. Option 1. Diversion (2) Path 1. Divert 50% and Three Material Streams (1 point) OR Path 2. Divert 75% and Four Material Streams (2 points) - Recommend OR Option 2. Reduction of Total Waste Material (2)

Credit/ Prerequisite	Points	Requirement
	Y/?/N	
		Recommend diverting at least 75% of the total construction and demolition material; diverted materials must include at least four material streams.

7.6 Indoor Environmental Quality: (Total: 16 points, suggested: 6 points)

Credit/ Prerequisite	Points Y/?/N	Requirement
7.7.1 Minimum Indoor Air Quality Performance		
Prerequisite		Must: Meet the LEED requirements for both ventilation and monitoring. Recommend 1. Design outside air flow to meet the requirement of ASHRAE 62.1–2010 or a local equivalent. 2. Install flow meter or flow switch to monitor outside air flow, alarm is required if out of spec. 3. Install CO2 sensor for natural ventilation area.
7.7.2 Environmental Tobacco Smoke Control		
Prerequisite		Must: Prohibit smoking inside the building. Prohibit smoking outside the building except in designated smoking areas located at least 25 feet (7.5 meters) from all entries, outdoor air intakes, and operable windows. Also prohibit smoking outside the property line in spaces used for business purposes. Signage must be posted within 10 feet (3 meters) of all building entrances indicating the no-smoking policy.
7.7.3 Enhanced Indoor Air Quality Strategies		
Credit	2/0/0	Option 1. Enhanced IAQ Strategies (1 point) Option 2. Additional Enhanced IAQ Strategies (1 point) Recommend 1. Install permanent entryway systems at least 10 feet (3 meters) long in the primary direction of travel to capture dirt and particulates entering the building at regularly used exterior entrances. 2. each space where hazardous gases or chemicals may be present or used (e.g., garages, housekeeping and laundry areas, copying and printing rooms), to create negative pressure with respect to adjacent spaces when the doors to the room are closed. For each of these spaces, provide self-closing doors and deck-to-deck partitions or a hard-lid ceiling. 3. minimum efficiency reporting value (MERV) of 13 or higher, in accordance with ASHRAE Standard 52.2–2007; or Class F7 or higher as defined by CEN Standard EN 779–2002 required for outside air intake system. 4. CO2 sensor required for densely(25 p/93m2) occupied spaces. As v4.1 is more flexible, if above is not easy to achieve, team can try v4.1 Comply with 3 strategies for 1 point or 6 strategies for 2 points Strategy 1. Entryway Systems Strategy 2. Interior Cross-Contamination Prevention Strategy 3. Filtration of Outdoor Air

Credit/ Prerequisite	Points	Requirement
	Y/?/N	
		Strategy 4. Filtration of Recirculated Air Strategy 5. Increased Ventilation 15 Percent Strategy 6. Increased Ventilation 30 Percent Strategy 7. Operable Windows Strategy 8. Engineered Natural Ventilation Strategy 9. Carbon Dioxide Monitoring Strategy 10. Additional Source Control and Monitoring
7.7.4 Low-Emitting Materials		
Credit	1/1/1	This credit includes requirements for product manufacturing as well as project teams. It covers volatile organic compound (VOC) emissions in the indoor air and the VOC content of materials, as well as the testing methods by which indoor VOC emissions are determined. Option 1. Product Category Calculations (3) or Option 2. Budget Calculation Method (3) Recommend interior paints and coatings, interior adhesives and sealants, flooring, ceilings, walls, thermal and acoustic insulation use low-emitting material.
7.7.5 Construction Indoor Air Quality Management Plan		
Credit	1/0/0	Develop and implement an indoor air quality (IAQ) management plan for the construction and preoccupancy phases of the building. Recommend 1. During construction, meet or exceed all applicable recommended control measures of the Sheet Metal and Air Conditioning National Contractors Association (SMACNA) IAQ Guidelines for Occupied Buildings under Construction, 2nd edition, 2007, ANSI/SMACNA 008–2008, Chapter 3. 2. Do not operate permanently installed air-handling equipment during construction unless filtration media with a minimum efficiency reporting value (MERV) of 8 (or equivalent filtration media class of F5 or higher), are installed at each return air grille and return or transfer duct inlet opening such that there is no bypass around the filtration media.
7.7.6 Indoor Air Quality Assessment		
Credit	1/1/0	Select one of the following two options, to be implemented after construction ends and the building has been completely cleaned. Option 1. Flush-Out (1 point) - Recommend OR Option 2. Air Testing (2 points) Recommend performing a building flush-out by supplying a total air volume of 14,000 cubic feet of outdoor air per square foot (4,267,140 liters of outdoor air per square meter) of gross floor area while maintaining an internal temperature of at least 60°F (15°C) and no higher than 80°F (27°C) and relative humidity no higher than 60%.

Credit/ Prerequisite	Points	Requirement																																											
	Y/?/N																																												
		If possible, conducting baseline IAQ testing after construction ends and before occupancy, but under ventilation conditions typical for occupancy, to meet option 2. As v4.1 option 2 is flexible, team can try it if above is hard to achieve. Projects can now select the type of contaminants to be tested in form of organic or inorganic compounds (1 point each) or both (2 points) via split pathways for particulate matter (PM) and organic gases testing and/or volatile organic compound (VOC) testing. Path 1. Particulate Matter and Inorganic Gases (1 point) <table><tr><th>Contaminant (CAS#)</th><th>Concentration Limit (µg/m³)</th><th>Allowed Test Methods</th></tr><tr><td>Carbon monoxide (CO)</td><td>9 ppm; no more than 2 ppm above outdoor levels</td><td>ISO 4224 EPA Compendium Method IP-3 GB/T 18883-2002 for projects in China Direct calibrated electrochemical instrument with accuracy of +/- 3% of reading and resolution of 0.1 ppm NDIR CO Sensors with accuracy of 1% of 10 ppm full scale and display resolution of less than 0.1ppm</td></tr><tr><td>PM 10</td><td>ISO 14644-1:2015, cleanroom class of 8 or lower 50 µg/m³ Healthcare only: 20 µg/m³</td><td rowspan="2">Particulate monitoring device with accuracy greater of 5 micrograms/m³ or 20% of reading and resolution (5 min average data) +/- 5 µg/m³</td></tr><tr><td>PM 2.5</td><td>12 µg/m³ or 35 µg/m³**</td></tr><tr><td>Ozone</td><td>0.07 ppm</td><td>Monitoring device with accuracy greater of 5 ppb or 20% of reading and resolution (5 min average data) +/- 5 ppb ISO 13964 ASTM D5149 -- O2 EPA designated methods for Ozone</td></tr></table> AND/OR Path 2. Volatile Organic Compounds (1 point) <table><tr><th>Contaminant (CAS#)</th><th>Concentration Limit (µg/m³)</th><th>Allowed Test Methods</th></tr><tr><td>Formaldehyde 50-00-0</td><td>20 µg/m³ (16 ppb)</td><td rowspan="2">ISO 16000-3, 4; EPA TO-11a, EPA comp. IP-6A ASTM D5197-16</td></tr><tr><td>Acetaldehyde 75-07-0</td><td>140 µg/m³</td></tr><tr><td>Benzene 71-43-2</td><td>3 µg/m³</td><td rowspan="10">ISO 16000-6 EPA IP-1, EPA TO-17, EPA TO-15 ISO 16017-1, 2; ASTM D6196-15</td></tr><tr><td>Hexane (n-) 110-54-3</td><td>7000 µg/m³</td></tr><tr><td>Naphthalene 91-20-3</td><td>9 µg/m³</td></tr><tr><td>Phenol 108-95-2</td><td>200 µg/m³</td></tr><tr><td>Styrene 100-42-5</td><td>900 µg/m³</td></tr><tr><td>Tetrachloroethylene 127-18-4</td><td>35 µg/m³</td></tr><tr><td>Toluene 108-88-3</td><td>300 µg/m³</td></tr><tr><td>Vinyl acetate 108-05-4</td><td>200 µg/m³</td></tr><tr><td>Dichlorobenzene (1,4-) 106-46-7</td><td>800 µg/m³</td></tr><tr><td>Xylenes-total 108-38-3, 95-47-6, and 106-42-3</td><td>700 µg/m³</td></tr></table>	Contaminant (CAS#)	Concentration Limit (µg/m³)	Allowed Test Methods	Carbon monoxide (CO)	9 ppm; no more than 2 ppm above outdoor levels	ISO 4224 EPA Compendium Method IP-3 GB/T 18883-2002 for projects in China Direct calibrated electrochemical instrument with accuracy of +/- 3% of reading and resolution of 0.1 ppm NDIR CO Sensors with accuracy of 1% of 10 ppm full scale and display resolution of less than 0.1ppm	PM 10	ISO 14644-1:2015, cleanroom class of 8 or lower 50 µg/m³ Healthcare only: 20 µg/m³	Particulate monitoring device with accuracy greater of 5 micrograms/m³ or 20% of reading and resolution (5 min average data) +/- 5 µg/m³	PM 2.5	12 µg/m³ or 35 µg/m³**	Ozone	0.07 ppm	Monitoring device with accuracy greater of 5 ppb or 20% of reading and resolution (5 min average data) +/- 5 ppb ISO 13964 ASTM D5149 -- O2 EPA designated methods for Ozone	Contaminant (CAS#)	Concentration Limit (µg/m³)	Allowed Test Methods	Formaldehyde 50-00-0	20 µg/m³ (16 ppb)	ISO 16000-3, 4; EPA TO-11a, EPA comp. IP-6A ASTM D5197-16	Acetaldehyde 75-07-0	140 µg/m³	Benzene 71-43-2	3 µg/m³	ISO 16000-6 EPA IP-1, EPA TO-17, EPA TO-15 ISO 16017-1, 2; ASTM D6196-15	Hexane (n-) 110-54-3	7000 µg/m³	Naphthalene 91-20-3	9 µg/m³	Phenol 108-95-2	200 µg/m³	Styrene 100-42-5	900 µg/m³	Tetrachloroethylene 127-18-4	35 µg/m³	Toluene 108-88-3	300 µg/m³	Vinyl acetate 108-05-4	200 µg/m³	Dichlorobenzene (1,4-) 106-46-7	800 µg/m³	Xylenes-total 108-38-3, 95-47-6, and 106-42-3	700 µg/m³
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Naphthalene 91-20-3	9 µg/m³																																												
Phenol 108-95-2	200 µg/m³																																												
Styrene 100-42-5	900 µg/m³																																												
Tetrachloroethylene 127-18-4	35 µg/m³																																												
Toluene 108-88-3	300 µg/m³																																												
Vinyl acetate 108-05-4	200 µg/m³																																												
Dichlorobenzene (1,4-) 106-46-7	800 µg/m³																																												
Xylenes-total 108-38-3, 95-47-6, and 106-42-3	700 µg/m³																																												

Credit/ Prerequisite	Points Y/?/N	Requirement
7.7.7 Thermal Comfort		
Credit	0/0/1	To meet the Thermal Comfort Design and Thermal Comfort Control Thermal Comfort Design Option 1. ASHRAE Standard 55-2010 OR Option 2. ISO and CEN Standards Thermal Comfort Control (Cannot meet for FAB) Provide individual thermal comfort controls for at least 50% of individual occupant spaces. Provide group thermal comfort controls for all shared multioccupant spaces. Thermal comfort controls allow occupants, whether in individual spaces or shared multioccupant spaces, to adjust at least one of the following in their local environment: air temperature, radiant temperature, air speed, and humidity.
7.7.8 Interior Lighting		
Credit	1/1/0	Select one or both of the following two options. Option 1. Lighting Control (1 point) AND/OR Option 2. Lighting Quality (1 point) Recommend 1. use light fixtures with a luminance of less than 2,500 cd/m2 between 45 and 90 degrees from nadir. 2. use light sources with a CRI (Color Rendering Index) of 80 or higher 3. use light sources that have a rated life (or L70 for LED sources) of at least 24,000 hours (at 3-hour per start, if applicable). 4. Use direct-only overhead lighting for 25% or less of the total connected lighting load for all regularly occupied spaces. 5. For at least 90% of the regularly occupied floor area, meet or exceed the following thresholds for area-weighted average surface reflectance: 85% for ceilings, 60% for walls, and 25% for floors.
7.7.10 Daylight		
Credit	0/0/3	(FAB Cannot meet) Provide manual or automatic (with manual override) glare-control devices for all regularly occupied spaces. Option 1. Simulation: Spatial Daylight Autonomy and Annual Sunlight Exposure (3) OR Option 2. Simulation: Illuminance Calculations (2) OR Option 3. Measurement (3)
7.7.11 Quality Views		
Credit	0/0/1	(FAB Cannot meet) Achieve a direct line of sight to the outdoors via vision glazing for 75% of all regularly occupied floor area.

Credit/ Prerequisite	Points	Requirement
	Y/?/N	
7.7.12 Acoustic Performance		
Credit	0/0/1	(Very difficult to meet) For all occupied spaces, meet the following requirements, as applicable, for HVAC background noise, sound isolation, reverberation time, and sound reinforcement and masking.

7.7 Innovation: (Total: 6 points, suggested: 6 points)

Credit/ Prerequisite	Points	Requirement
	Y/?/N	
7.8.1 Innovation		
Credit	5/0/0	Recommend 1. Non mercury LED lighting 2. Establish on-site signage system and navigation plan 3. Write a case study 4. Share this case on the website
7.8.2 LEED Accredited Professional		
Credit	1/0/0	Recommend At least one principal participant of the project team must be a LEED Accredited Professional (AP) with a specialty appropriate for the project.

7.8 Regional Priority: (Total: 4 points, suggested: 4 points)

Credit/ Prerequisite	Points	Requirement
	Y/?/N	
7.9.1 Reginal Priority		
Credit	4/0/0	Recommend reach the item below to get Regional Priority point Rain water management Outdoor water use reduction Indoor water use reduction Above is common item for TWN, SGP, MAS, JPN. Enhanced indoor air quality strategies (TWN, SGP, MAS) Enhanced commissioning (JPN) Other area please Refer LEED v4 for BUILDING DESIGN AND CONSTRUCTION for the detail.

8.0 Document Control

8.1 **Approval:** GLOBAL_FAC_GFTT_DOC_APPR

8.2 Notification:

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
GLOBAL_FAC_NOTIFY
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY

8.3 Final Viewer Access

All Micron Team Members from all Micron Sites

8.4 Retention

Adherence to the requirements specified in the
[Global Facilities - Document Control Procedures](#)

8.5 Review

Biennially (two years)

9.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New document	08/31/21
1	7.0	Updated LEED Gold Strategy - Details and Recommendations to obtain LEED Points	08/15/23

Appendix 1: Template Form for Site Specific Deviation List

Site: _____

No	Specification Section and Item	Deviation Proposal	Scope Impact (Benefit, Cost, Schedule)	Remark	Approval: 1. Site 2. GFTT